

Statement of Work – Drill Engineering and Maintenance Support
February 2025

Effective date: 15 February 2025

End date: 31 March 2026



Quote CLIN	Task	2/1/25 – 3/31/26 (PY25)
3	TRIDENT Drill Maintenance and Support (a) Field support during Arctic impact crater drill field tests. (b) Virtual field support for remote drilling validation and tests. (c) Provide Honeybee drill SW updates as necessary. (d) Support field readiness review and periodic technical consultations and meetings. (e) Travel costs for one field engineer to Arctic test site. (f) Various small fasteners, fixtures and materials incidental and necessary for TRIDENT support.	
1	TRIDENT Drill Repair (a) An assessment of drill mechanism health from visual inspection for any/all issues, follow-up to prior (August 2024) field assessment. This requires a head-end teardown and disassembly. (b) Motor(s) and mounts and other head-end assemblies will be removed and replaced as necessary, either with an identical replacement or a retrofitted unit of similar operational characteristics. (c) The reassembled head-end unit will be verified and validated to perform to design specifications both mechanically and with the drill avionics. (d) The drill will be exercised with a test sample to ensure a proper repair has been performed.	
2	New TRIDENT Drill Test Stand	

	(a) Design and fabricate a drill stand for the NASA Ames (PR1) TRIDENT drill. (b) Drill stand design will accommodate standalone operation on loose soil and surfaces as well as anchoring to a floor or test chamber frame. (c) Drill stand will be optimized for safe packaging and transportation to/in remote drill sites, including by air cargo, truck and manual carrying. (d) Where feasible, the drill stand will be designed from COTS aluminum extrusions and mounting hardware that is readily and commonly available. (e) As-built bolt/fastener patterns, dimensions and type information will be provided to facilitate future field spares.	
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